Technology Stack (Architecture & Stack)

Project Detail	Information
Date	29 October 2026
Team ID / Enrollment No.	EN25CS3010450
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	2 Marks
Student	Joyal Sengar (Individual Project)

Define Your Technology Stack:

Identify the technologies, platforms, data sources and BI tools used to build the analytical solution. The selected stack should be suitable for preparing, analyzing and presenting the global fuel prices and taxation dataset.

S.No.	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Language or calculation logic used to filter, transform, aggregate and analyze data.	Tableau calculated fields and aggregations	Used to calculate or summarize fuel-price measures and compare petrol, diesel, LPG, tax percentage and Brent crude price across available countries, regions, dates, income levels and subsidy levels.
2	Database / Data Warehouse	Storage location for structured data used in analysis and reporting.	CSV dataset (global fuel prices dataset)	Stores the available country, region, date, fuel-price, Brent crude price, tax, income-level and subsidy-level fields. Tableau can connect directly to the CSV without requiring a separate database server.

Technology Stack (continued)

S.No.	Component	Description	Technology	Justification / Purpose
3	Data Ingestion	Method used to bring source data into the analytics environment.	Tableau Desktop data source connection (Text File / CSV)	Loads the fuel dataset into Tableau for inspection and visualization. The source file can be reconnected or refreshed when an updated file is available.
4	Data Transformation and Business Logic	Tools and logic used to clean, organize, filter and aggregate data for analysis.	Tableau Desktop: Data Source tab, field data types, filters, aggregations and calculated fields	Supports checking field types, date formatting, missing or duplicate records, category consistency and unusual values, then preparing the measures needed for comparison. Any cleaning decisions should be documented.
5	Data Visualization / Analytics	Tools used to create charts, reports and visual analysis.	Tableau Desktop worksheets: bar charts, line charts, KPI-style summaries, heatmaps and other views used in the workbook	Presents comparisons of fuel prices by country and region, patterns over time, tax-percentage differences, and comparisons by income and subsidy categories, according to the views included in the workbook.
6	Front-End / User Interface	Interface through which users view and interact with the analytical results.	Tableau dashboard and story in the packaged Tableau workbook (.twbx)	Combines the worksheets into a dashboard and data story for easier interpretation. Filters, highlighting, tooltips or online sharing should be described as available only where they are actually configured in the submitted workbook.

Implementation note: This stack reflects the CSV-based Tableau workflow described for the project. It does not assume a separate SQL database, programming language, automatic refresh schedule or Tableau Public publication. Include those only if you actually used or configured them.